



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Vo Lam Quan
15 Dec 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

Phase	Key Methodology	Description
Preparation	Imputation & Encoding	Cleaned data by filling missing values (e.g., using Mode) and converted categorical features (e.g., Launch Site) to numeric using One-Hot Encoding .
Model Optimization	GridSearchCV	Systematically tuned the hyperparameters of each model to find the absolute best settings (e.g., finding the optimal n_neighbors for KNN).
Model Development	4 Classification Models	Compared Logistic Regression, SVM, K-Nearest Neighbors, and Decision Tree to find the most suitable algorithm.
Evaluation	Metric Comparison	All final, optimized models were scored using Accuracy, Precision, and Recall on a reserved Test Set .

Summary of all results

Model	Highest Performance Metric	Why It Matters for Landing
KNN	Highest Precision	Best for Safety: Lowest rate of predicting success when failure occurs.
Logistic Regression & SVM	Highest Accuracy	Best for Efficiency: Most successful at catching true successful landings.
Decision Tree	Moderate/Lowest	Used as a comparative baseline.

Introduction

- **Project background and context**
 - **Project Goal:** To build and test machine learning models that accurately predict the success or failure of a rocket's first stage landing (Landed vs. Not Landed).
 - **Significance:** Accurate prediction is critical for safety and cost-efficiency. It allows missions to pre-position recovery assets only when success is highly probable, saving millions.
 - **Data:** Historical launch records containing features like Payload Mass, Launch Site, and Orbit Type.
- **Problems you want to find answers**
 - Which Model is Best?
 - Find the model with the highest overall predictive power on unseen data (highest AUC-ROC).
 - What is the Biggest Risk?
 - Determine which model has the highest Precision to minimize the costliest error: predicting a success when it was actually a failure (False Positive).
 - What Drives Success?
 - Identify the most important launch parameters that influence a successful landing (using feature importance).

Section 1

Methodology

Methodology

Executive Summary

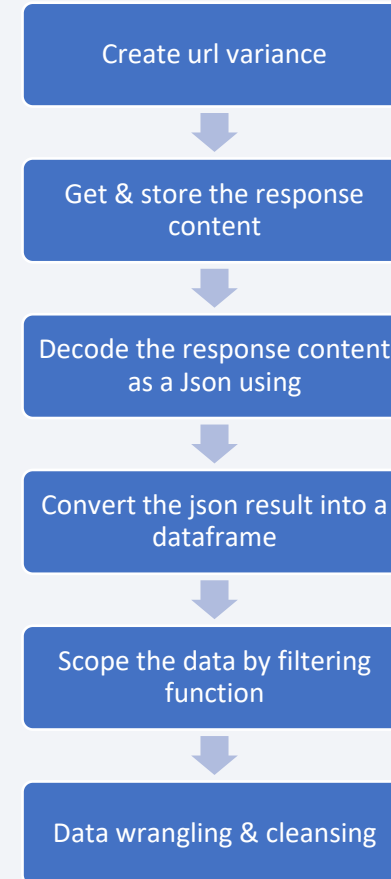
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

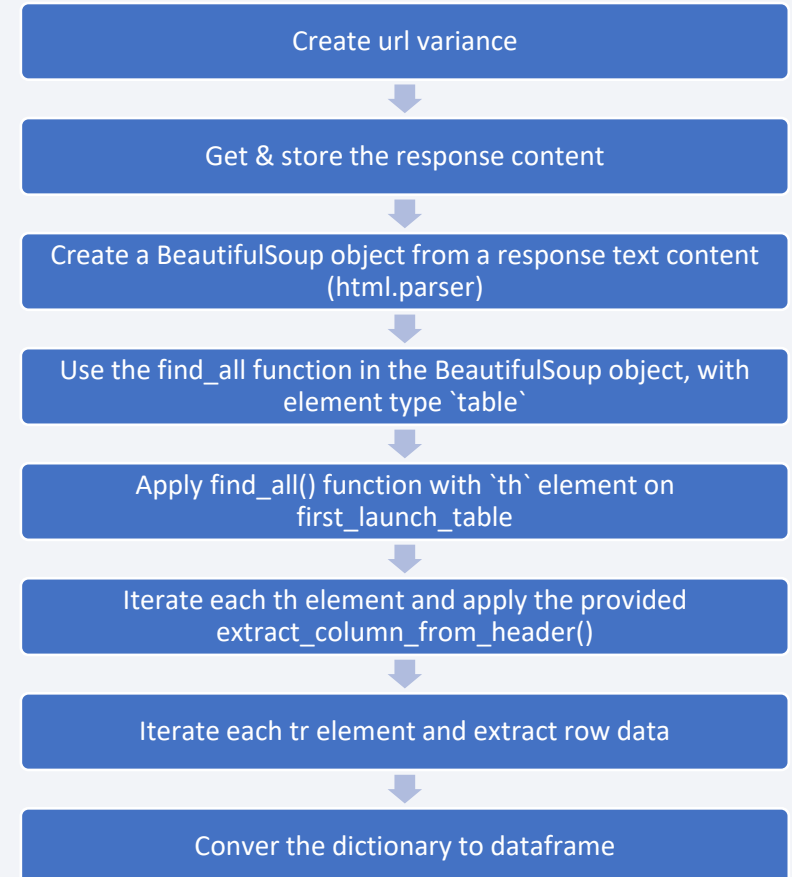
Data Collection – SpaceX API

- Key Steps to get the data from `api.spacexdata.com`
 - Use request, get to download data api data from web, result is in json format.
 - Decode with JSON and covert json into dataframe with `JSON_NORMALIZE`
- Data Wrangling & Cleansing
 - Profiling data and detect null, blank & missing data.
 - Handling missing data depending on the types.
 - Filter only relevant data.
- [Attached GitHub URL of the completed SpaceX API calls notebook](#)



Data Collection - Scraping

- Key Steps to download Falcon 9 Launch data from en.wikipedia.org
 - Use request, get to download data api data from web, result is in html format.
 - Use BeautifulSoup to parse the data and store in a text object.
- Extract data from html format
 - Navigating to the element “table”, and select the position of table we want to extract as we already know what are they.
 - Navigating to the element “th”, we can identify all columns of tables.
 - Create a dictionary with above columns, looping all the row by the element “tr” to extract data & store in the dictionary
 - Create a dataframe from the above dictionary
- [Attached GitHub URL of the completed web scraping notebook](#)



Data Wrangling

- Check the data types
- Count number of launches per each launch site
- Count number and occurrence of each orbit
- Count number and each landing_outcomes
- Create a new Class (0 for bad outcomes and 1 for remaining outcomes), bad_outcomes is a list of selected landing_outcomes that not succeed.
- [Attached GitHub URL of the completed data wrangling notebook](#)

Use the method `value_counts()` on the column `LaunchSite`



Use the method `.value_counts()` to determine the number and occurrence of each orbit in the column `Orbit`



Use the method `.value_counts()` on the column `Outcome` to determine the number of landing_outcomes. Then assign it to a variable `landing_outcomes`.



Create a list `bad_outcomes` to store all the not succeed ones



Create new Class columns derived from the `bad_outcomes` one
`landing_class` = 0 if `bad_outcome`
`landing_class` = 1 otherwise

EDA with Data Visualization

1. Visualize the relationship between Flight Number, Launch Site, and Class with **catplot** to understand whether the number of flight has any impact to land or not landed against different launch sites
2. Visualize the relationship between Payload Mass and Launch Site with **catplot** to understand whether the payload has any impact to land or not landed against different launch sites
3. Visualize the relationship between success rate of each orbit type with **barplot** to understand which orbit types have better success rate and which ones have less
4. Visualize the relationship between Flight Number and Orbit type with **scatterplot** to understand whether more flights increase the chance of land against each orbit type or not
5. Visualize the relationship between Payload Mass and Orbit type on Y while Flight Number on X with **scatterplot** to understand at more heavy payloads success, what orbit has more success and which ones are not.
6. Visualize the launch success yearly trend with **lineplot**

Note: After this, the function `get_dummies` is used to apply **OneHotEncoder** to the column Orbits, LaunchSite, LandingPad, and Serial. Then we cast all numeric columns into float64 with `astype('float64')`

[Attached GitHub URL of the completed EDA with Visualization notebook](#)

EDA with SQL

- All Launch Site Names: using distinct to find all launch site names.
- Launch Site Names Begin with 'CCA': using like to find launch site names begins with 'CCA', and limit only 5 records in result.
- Total Payload Mass: using where to restrict the 'NASA (CRS)' customer, sum to calculate the total payload and group by keyword to have sum by customer.
- Average Payload Mass by F9 v1.1: using where to restrict the 'F9 v1.1' booster_version, average to calculate the average of payloads and group by keyword to have average by booster_version.
- First Successful Ground Landing Date: using where to restrict the 'Success (ground pad)' Landing_Outcome, min to get the smallest launch date.
- Successful Drone Ship Landing with Payload between 4000 and 6000: using where to restrict the 'Success (drone ship)' Landing_Outcome, between/and to restrict the payload within range 4000-6000.
- Total Number of Successful and Failure Mission Outcomes: using count(*) to count all the launches and group by keyword to have the count by each Mission_Outcome.
- Boosters Carried Maximum Payload: using subquery with max keyword to get the max of payload, then pass this value into where clause to restrict launches carried this much payload.
- 2015 Launch Records: using substr to get the month, year, where clause to restrict by 2015 and landing_outcome failure (drone ship)
- Rank Landing Outcomes Between 2010-06-04 and 2017-03-20: using count(*) to count all the launches, between/and for date to restrict the launch date within the range '2010-06-04' and '2017-03-20' and Landing_Outcome in the list to restrict only concerning landing outcome, group by clause for the counting, and order by desc clause with count(*) for sorting by the result of count.
- [Attached GitHub URL of your completed EDA with SQL notebook](#)

Build an Interactive Map with Folium

- Created circle & marker for NASA Johnson Space Center, circle is highlighted area and popup showing the title when we clicked on the map, marker is an icon-like, we can create like a text icon, and it will be displayed in the map.
- Same concept as above, created another 4 circles & 4 markers for 4 launch sites and add to the map to highlight and display SpaceX launch sites.
- Added a marker cluster object to simplify the map, it will group many markers having the same coordinate and displayed as number, represented the number of markers in same coordinate, a.k.a number of launches.
- Same concept as above, this time marking icon_color by from the Class column for Success & Failure by Green or Red. For example: from the zoom out most, we see 46 in the east coast, drill-down we will see 7 at CCAFS SLC-40, it is 7 launches, when click further we can see that there are 3 successes & 4 failures.
- Now that we want to figure out the distance from launch site to places like: city, coastline. Adding, the mouse position, it will show the lat and long when the mouse is at a place, we then can capture the coordinate we want to measure.
- After that, applying the same concept as above, create an icon text to mark for the place we to measure the distance from launch site, for example: coastline. Then implement the polyline with coordinates are the launch site & the coastline point. Applying the same for railways, highway and city.
- [Attached GitHub URL of your completed interactive map with Folium map](#)

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

- Pie chart & drop-down list selection of launch site interaction:
 - Create a dropdownlist to choose either particular launch site or All launch sites.
 - In the callback function, provide the output as a pie chart, input as a selected site value.
 - Implement logic if selected value is All, return the pie chart with all launch sites, if the selected value is particular launch site, pass this filter to restrict data from dataframe.
 - With this, every time users choose either All sites or particular site, the pie chart will be refreshed with updated data.
- Scatter chart & drop-down list selection of launch site & range slicer payload mass interaction:
 - Create a range slider to choose from/to for payload mass by kg.
 - In the callback, provide the output as a scatter chart, input 1 as a selected site value, input 2 as a selected range value of payload mass.
 - Implement logic to pass selected site value and selected payload value to filter dataframe, data then feeds to scatter chart.
 - With this, every time users choose sites or payload mass, the scatter chart will be refreshed with updated data.
- [Attached the GitHub URL of your completed Plotly Dash lab](#)

Predictive Analysis (Classification)

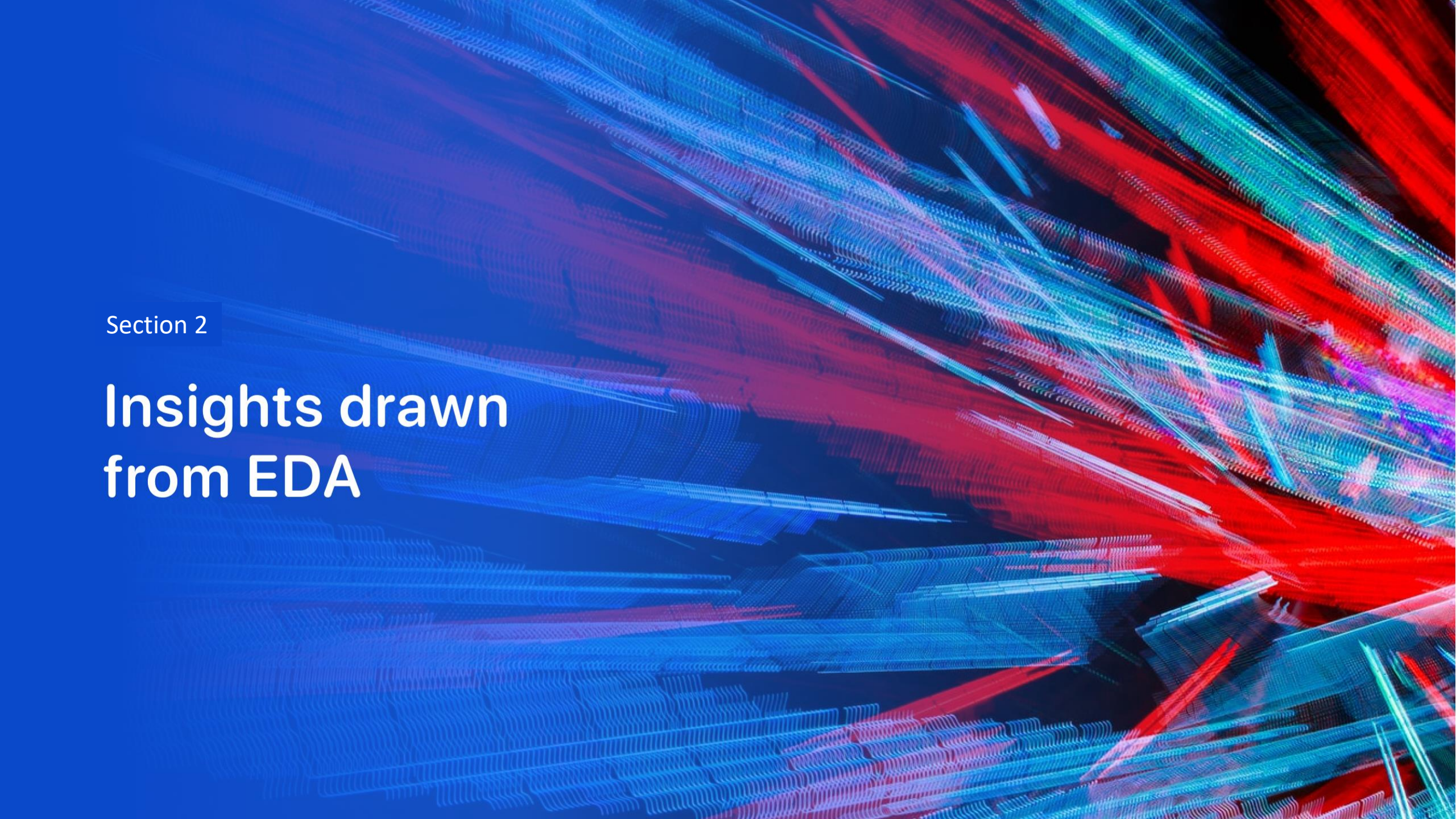
- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Prepressing data:
 - Standardize all the relevant data features.
 - Split the train, test datasets.
- Build: doing the below steps for 4 models: Logistic Regression, SVM, KNN and Decision Tree.
 - Create parameters, create grid search, fit the data to train the model, test the accuracy with test data
- Evaluate:
 - Compare all the results of these models, even the accuracy of LogReg & SVM is high, but considering the cost of false positive, select the model that perform the best at Precision: KNN
- [Attached the GitHub URL of your completed predictive analysis lab](#)

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

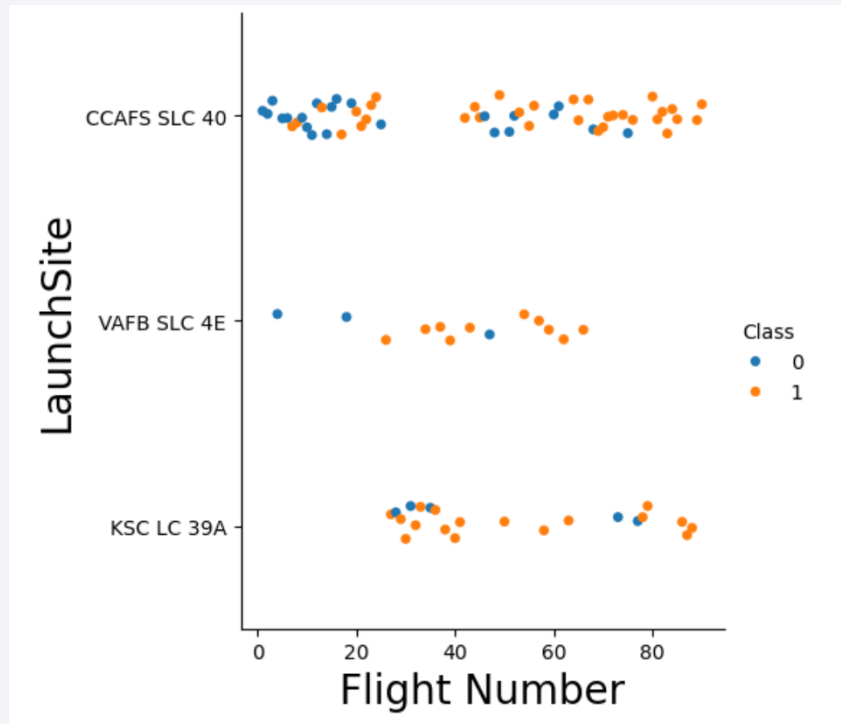
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

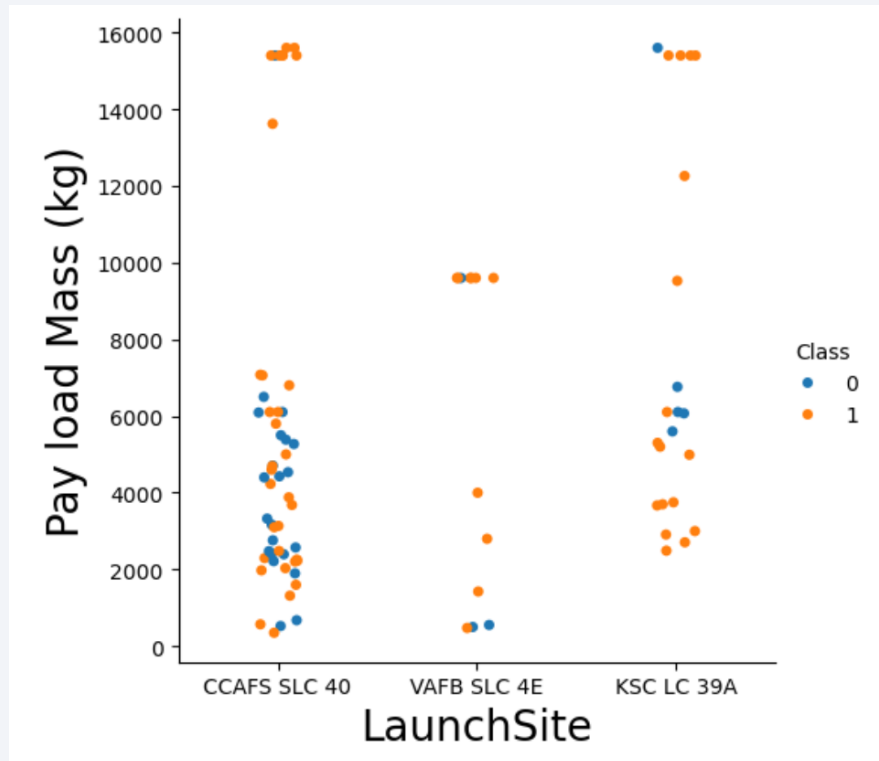
```
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the Launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect = 1)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("LaunchSite",fontsize=20)
plt.show()
```



- CCAFS SLC has the majority of launches, and the success of launches are significantly improved.
- Seem last 20 launches, there is no launch from VAFB SLC site at all.
- Number of launches used to happen a lot at KSC LC site, but not much recently, but most of them are succeed.

Payload vs. Launch Site

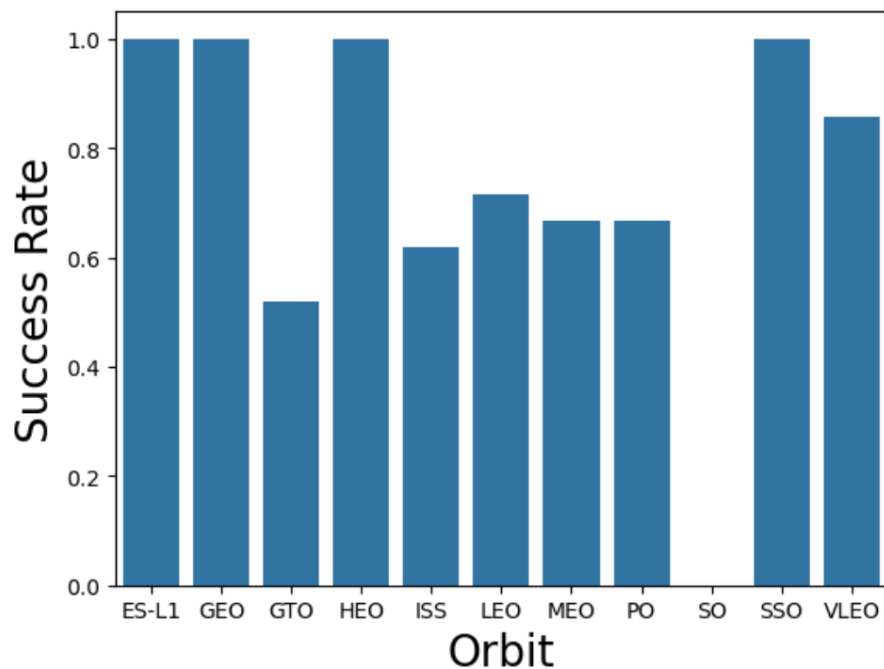
```
sns.catplot(y="PayloadMass", x="LaunchSite", hue="Class", data=df, aspect = 1)
plt.xlabel("LaunchSite",fontsize=20)
plt.ylabel("Pay load Mass (kg)",fontsize=20)
plt.show()
```



- For the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type

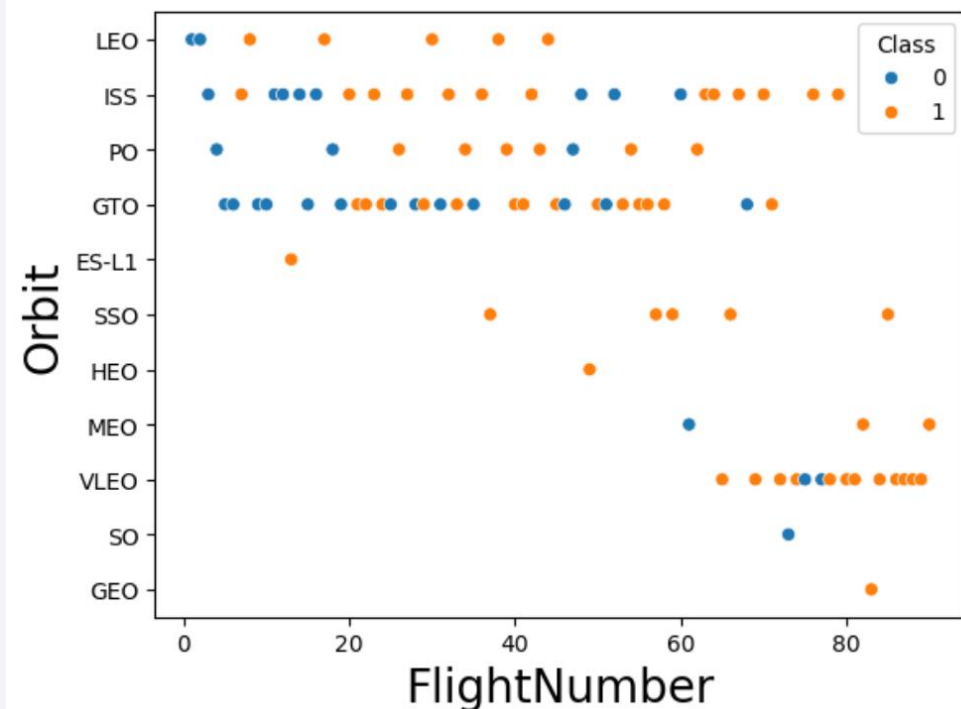
```
# HINT use groupby method on Orbit column and get the mean of Class column
grouped_mean = df.groupby('Orbit')['Class'].mean().to_frame(name='Class')
sns.barplot(y="Class", x="Orbit", data=grouped_mean)
plt.xlabel("Orbit", fontsize=20)
plt.ylabel("Success Rate", fontsize=20)
plt.show()
```



- ES-L1, GEO, HEO, SSO have the highest succeed rate at 100%
- GTO, ISS, MEO, PO are at bottom, success rate is around 50-65% only

Flight Number vs. Orbit Type

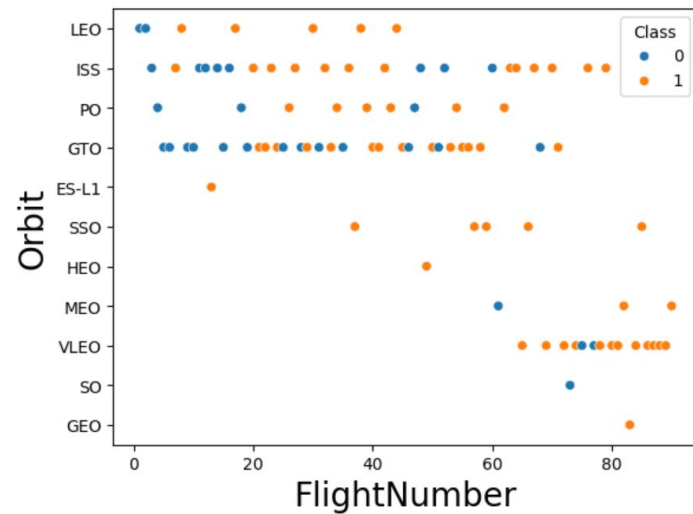
```
# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.scatterplot(y="Orbit", x="FlightNumber", hue="Class", data=df)
plt.xlabel("FlightNumber",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```



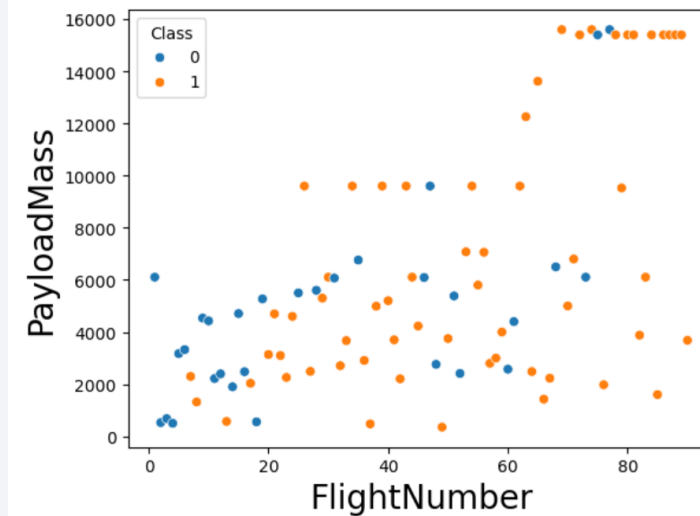
- In the LEO orbit, success seems to be related to the number of flights.
- Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

Payload vs. Orbit Type

```
# Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.scatterplot(y="Orbit", x="FlightNumber", hue="Class", data=df)
plt.xlabel("FlightNumber", fontsize=20)
plt.ylabel("Orbit", fontsize=20)
plt.show()
```



```
# Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.scatterplot(y="PayloadMass", x="FlightNumber", hue="Class", data=df)
plt.xlabel("FlightNumber", fontsize=20)
plt.ylabel("PayloadMass", fontsize=20)
plt.show()
```



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

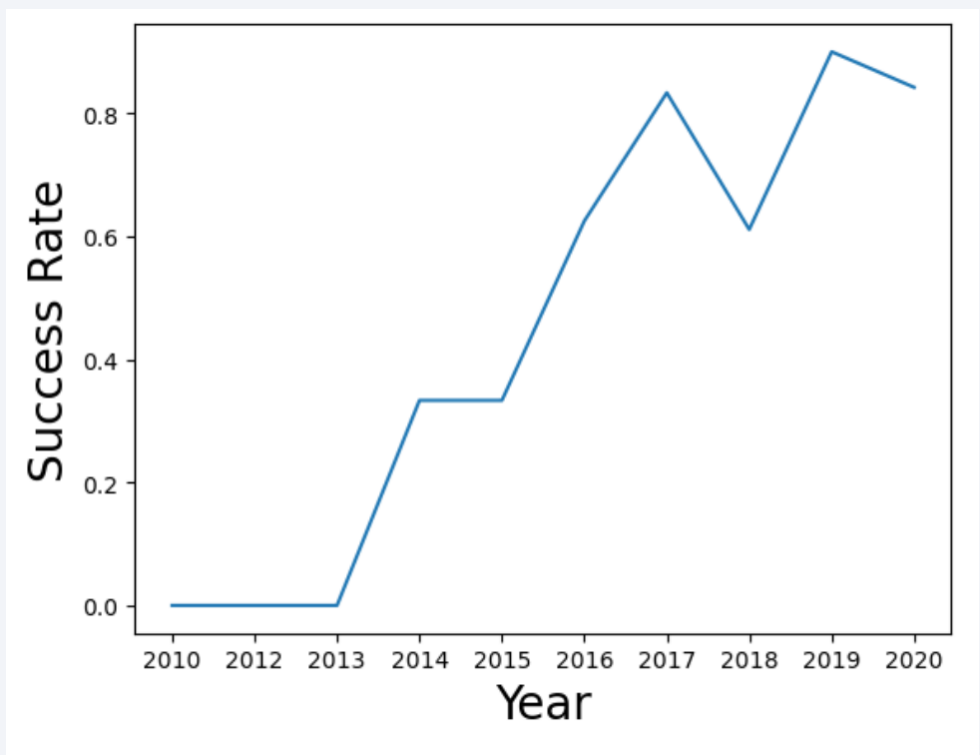
Launch Success Yearly Trend

```
# A function to Extract years from the date
year=[]
def Extract_year():
    for i in df["Date"]:
        year.append(i.split("-")[0])
    return year
Extract_year()
df['Date'] = year
df.head()
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins
0	1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False
1	2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False
2	3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False
3	4	2013	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False
4	5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False

```
# Plot a Line chart with x axis to be the extracted year and y axis to be the success rate
grouped_year_mean = df.groupby('Date')['Class'].mean().to_frame(name='Class')
sns.lineplot(y="Class", x="Date", data=grouped_year_mean)
plt.xlabel("Year",fontsize=20)
plt.ylabel("Success Rate",fontsize=20)
plt.show()
```

Visualize the launch success yearly trend with **lineplot**, clearly the success rate kept increasing since 2013 to 2020



All Launch Site Names

- Display the names of the unique launch sites in the space mission
- There are 4 launch sites

```
%sql select distinct Launch_Site from SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Display 5 records where launch sites begin with the string 'CCA'

```
%sql select * from SPACEXTABLE where Launch_Site like 'CCA%' LIMIT 5
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Display the total payload mass carried by boosters launched by NASA (CRS).
- 45596 kg is the total payload of all launches by customer NASA (CRS)

```
%sql select Customer,SUM(PAYLOAD_MASS__KG_) from SPACE_TABLE where Customer = 'NASA (CRS)' group by Customer
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Customer	SUM(PAYLOAD_MASS__KG_)
----------	------------------------

NASA (CRS)	45596
------------	-------

Average Payload Mass by F9 v1.1

- Display average payload mass carried by booster version F9 v1.1
- The F9 v1.1 booster version carries 2928.4 kg payload in average.

```
%sql select Booster_Version,AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE where Booster_Version = 'F9 v1.1' group by Booster_Version
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version	AVG(PAYLOAD_MASS__KG_)
-----------------	------------------------

F9 v1.1	2928.4
---------	--------

First Successful Ground Landing Date

- List the date when the first successful landing outcome in ground pad was achieved
- The first launch was on 22 Dec 2015 at CCAFS LC-40 site and carried 2034 kg payload.

```
%sql select min(date),* from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

min(date)	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2015-12-22	2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.
- Only 4 launches achieved this.

```
%sql select * from SPACEXTABLE where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS_KG_ between '4000' and '6000'
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2016-05-06	5:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-08-14	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
2017-10-11	22:53:00	F9 FT B1031.2	KSC LC-39A	SES-11 / EchoStar 105	5200	GTO	SES EchoStar	Success	Success (drone ship)

Total Number of Successful and Failure Mission Outcomes

- List the total number of successful and failure mission outcomes
- 100 success and only 1 failure

```
%sql select Mission_Outcome,count(*) from SPACEXTABLE group by Mission_Outcome
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
- There are 12 launches with the max payload of 15600 kg.

```
%sql select * from SPACEXTABLE where PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) from SPACEXTABLE)
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2019-11-11	14:56:00	F9 B5 B1048.4	CCAFS SLC-40	Starlink 1 v1.0, SpaceX CRS-19	15600	LEO	SpaceX	Success	Success
2020-01-07	2:33:00	F9 B5 B1049.4	CCAFS SLC-40	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600	LEO	SpaceX	Success	Success
2020-01-29	14:07:00	F9 B5 B1051.3	CCAFS SLC-40	Starlink 3 v1.0, Starlink 4 v1.0	15600	LEO	SpaceX	Success	Success
2020-02-17	15:05:00	F9 B5 B1056.4	CCAFS SLC-40	Starlink 4 v1.0, SpaceX CRS-20	15600	LEO	SpaceX	Success	Failure
2020-03-18	12:16:00	F9 B5 B1048.5	KSC LC-39A	Starlink 5 v1.0, Starlink 6 v1.0	15600	LEO	SpaceX	Success	Failure
2020-04-22	19:30:00	F9 B5 B1051.4	KSC LC-39A	Starlink 6 v1.0, Crew Dragon Demo-2	15600	LEO	SpaceX	Success	Success
2020-06-04	1:25:00	F9 B5 B1049.5	CCAFS SLC-40	Starlink 7 v1.0, Starlink 8 v1.0	15600	LEO	SpaceX, Planet Labs	Success	Success
2020-09-03	12:46:14	F9 B5 B1060.2	KSC LC-39A	Starlink 11 v1.0, Starlink 12 v1.0	15600	LEO	SpaceX	Success	Success
2020-10-06	11:29:34	F9 B5 B1058.3	KSC LC-39A	Starlink 12 v1.0, Starlink 13 v1.0	15600	LEO	SpaceX	Success	Success
2020-10-18	12:25:57	F9 B5 B1051.6	KSC LC-39A	Starlink 13 v1.0, Starlink 14 v1.0	15600	LEO	SpaceX	Success	Success
2020-10-24	15:31:34	F9 B5 B1060.3	CCAFS SLC-40	Starlink 14 v1.0, GPS III-04	15600	LEO	SpaceX	Success	Success
2020-11-25	2:13:00	F9 B5 B1049.7	CCAFS SLC-40	Starlink 15 v1.0, SpaceX CRS-21	15600	LEO	SpaceX	Success	Success

2015 Launch Records

- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- There are 2 failures drone ship in 2015

```
%sql select substr(Date, 6,2) as month,Date,Booster_Version,Launch_Site,Landing_Outcome from SPACEXTABLE where substr(Date,0,5)='2015' and Landing_Outcome = 'Failure (drone ship)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql select Landing_Outcome,count(*) from SPACEXTABLE where date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by count(*) desc
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	count(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

```
%sql select Landing_Outcome,count(*) from SPACEXTABLE where date between '2010-06-04' and '2017-03-20' and Landing_Outcome in ('Failure (drone ship)','Success (ground pad)') group by Landing_Outcome order by count(*) desc
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	count(*)
Failure (drone ship)	5
Success (ground pad)	3

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

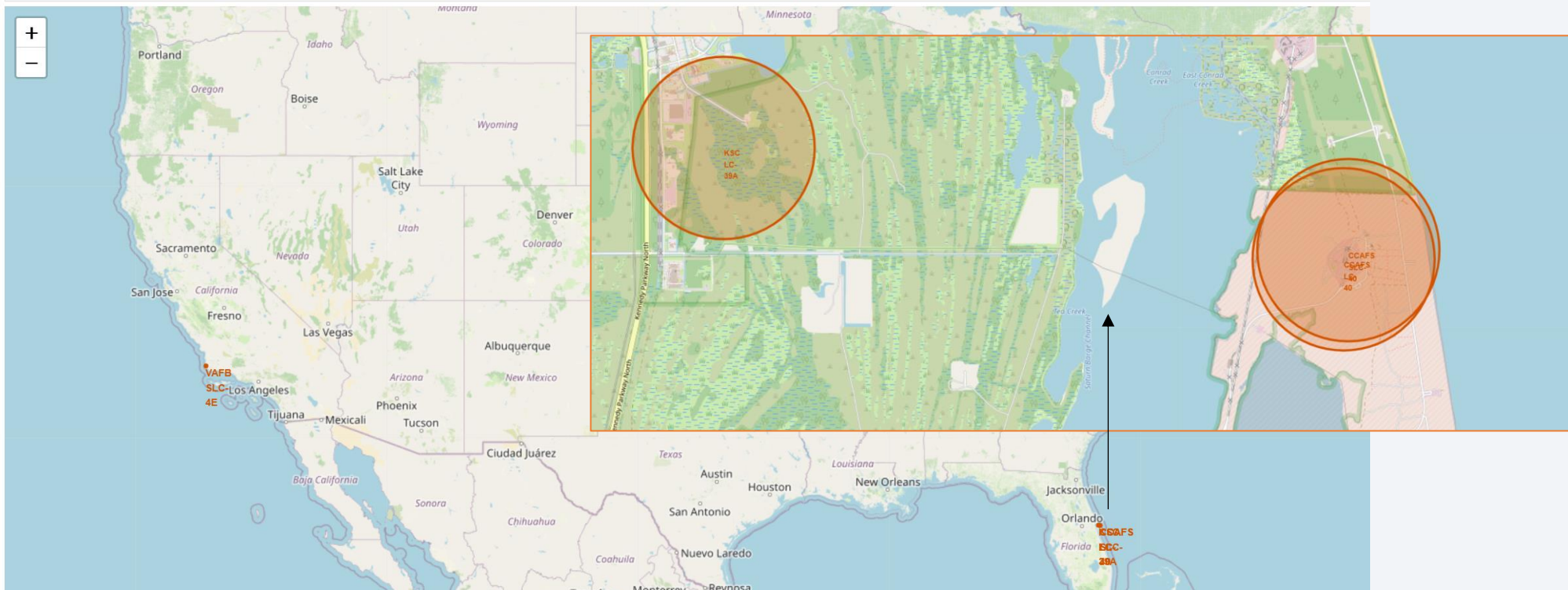
SpaceX launch sites in US

```
CCAFS_LC_coordinate = [28.562302, -80.577356]
CCAFS_SLC_coordinate = [28.563197, -80.576820]
KSC_coordinate = [28.573255, -80.646895]
VAFB_coordinates = [32.8156, -84.6174]

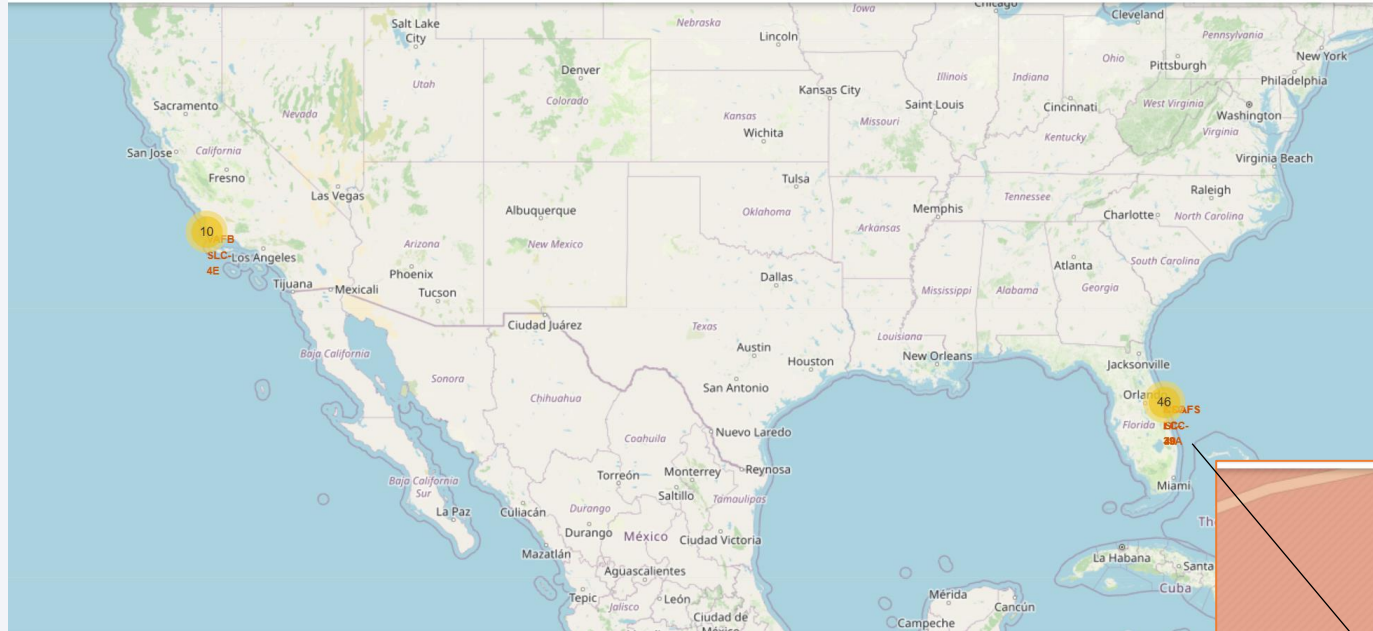
site_map.add_child(CCAFS_LC_circle)
site_map.add_child(CCAFS_LC_marker)
site_map.add_child(CCAFS_SLC_circle)
site_map.add_child(CCAFS_SLC_marker)
site_map.add_child(KSC_circle)
site_map.add_child(KSC_marker)
site_map.add_child(VAFB_circle)
site_map.add_child(VAFB_marker)
```

- All launch sites are in proximity to the Equator line
- All launch sites are in very close proximity to the coast; there are 1 in the west coast & 3 in the east coast.

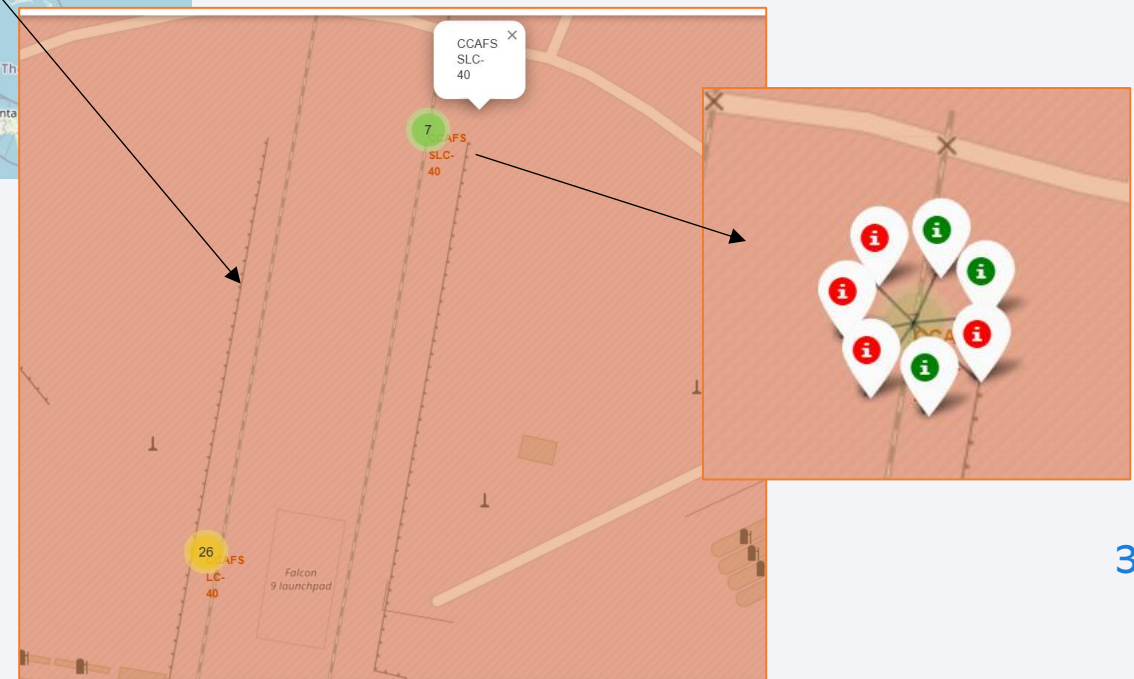
[10]:



Number of success & failed launches at each launch site



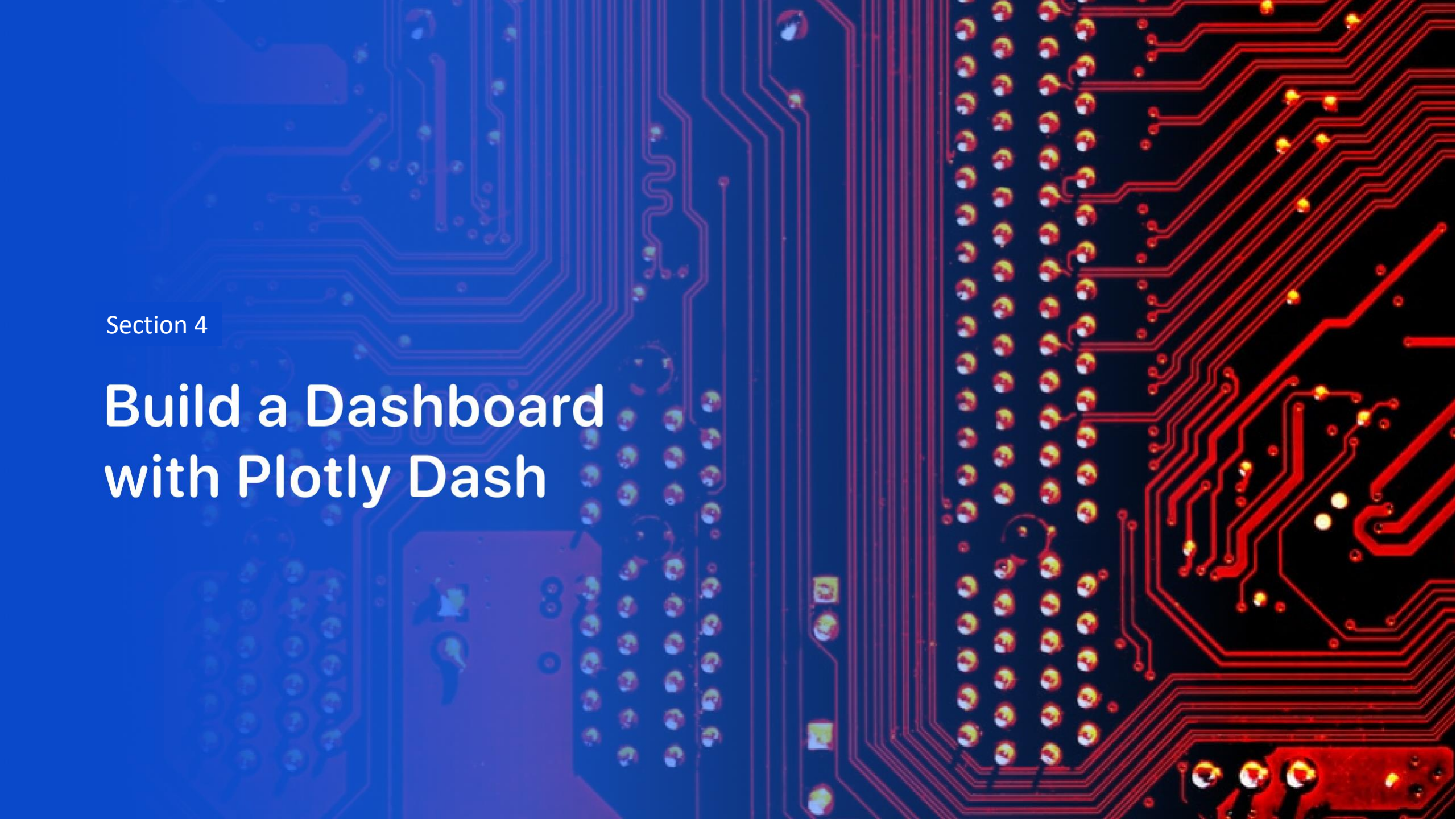
- With the number, we can easily identify how many launches in each launch site.
- Easily drill-down, with the color code, we can see how many success & failures. For example: at CCAFS SLC-40, there are 7 launches, when drill-down we can see that there are 3 successes & 4 failures



While launch site is near the railways, highways, coastline. They are kept very far from cities

- Closet coastline from launch site 'CCAFS SLC-40' is 0,86 km
- Closet railways from launch site 'CCAFS SLC-40' is 1.28 km
- Closet highways from launch site 'CCAFS SLC-40' is 0,58 km
- Closet city from launch site 'CCAFS SLC-40' is 51.43 km

→ Clearly the launch site is kept very far from the cities.



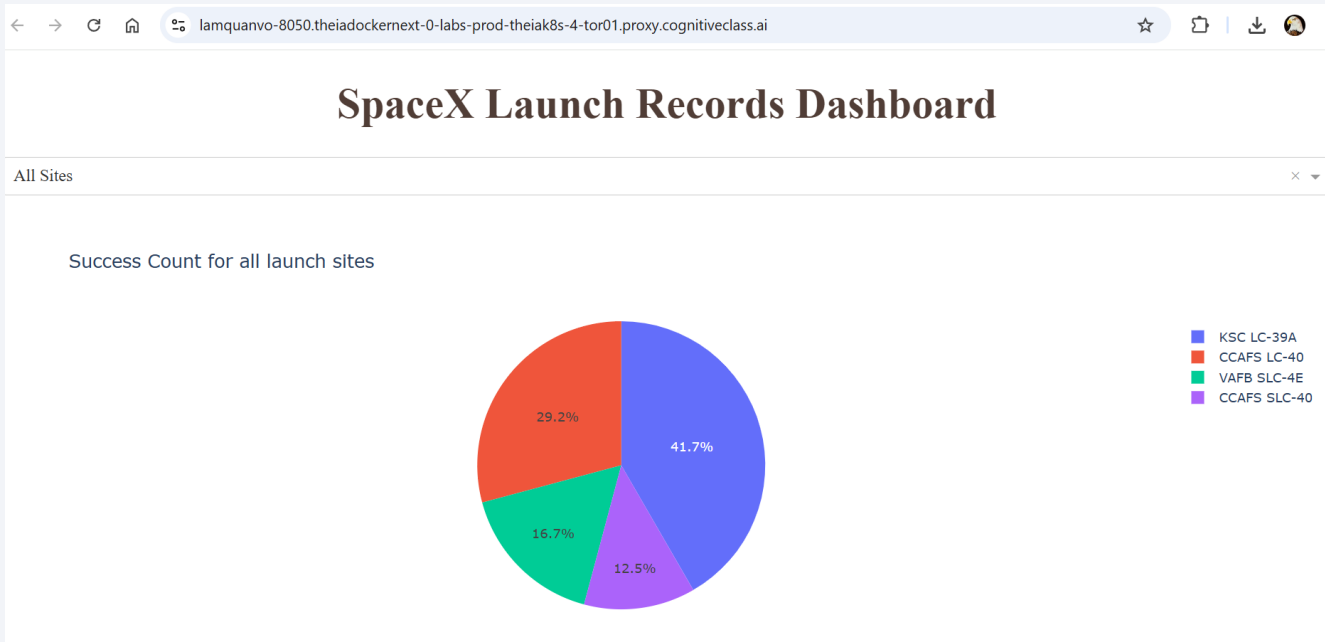
Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

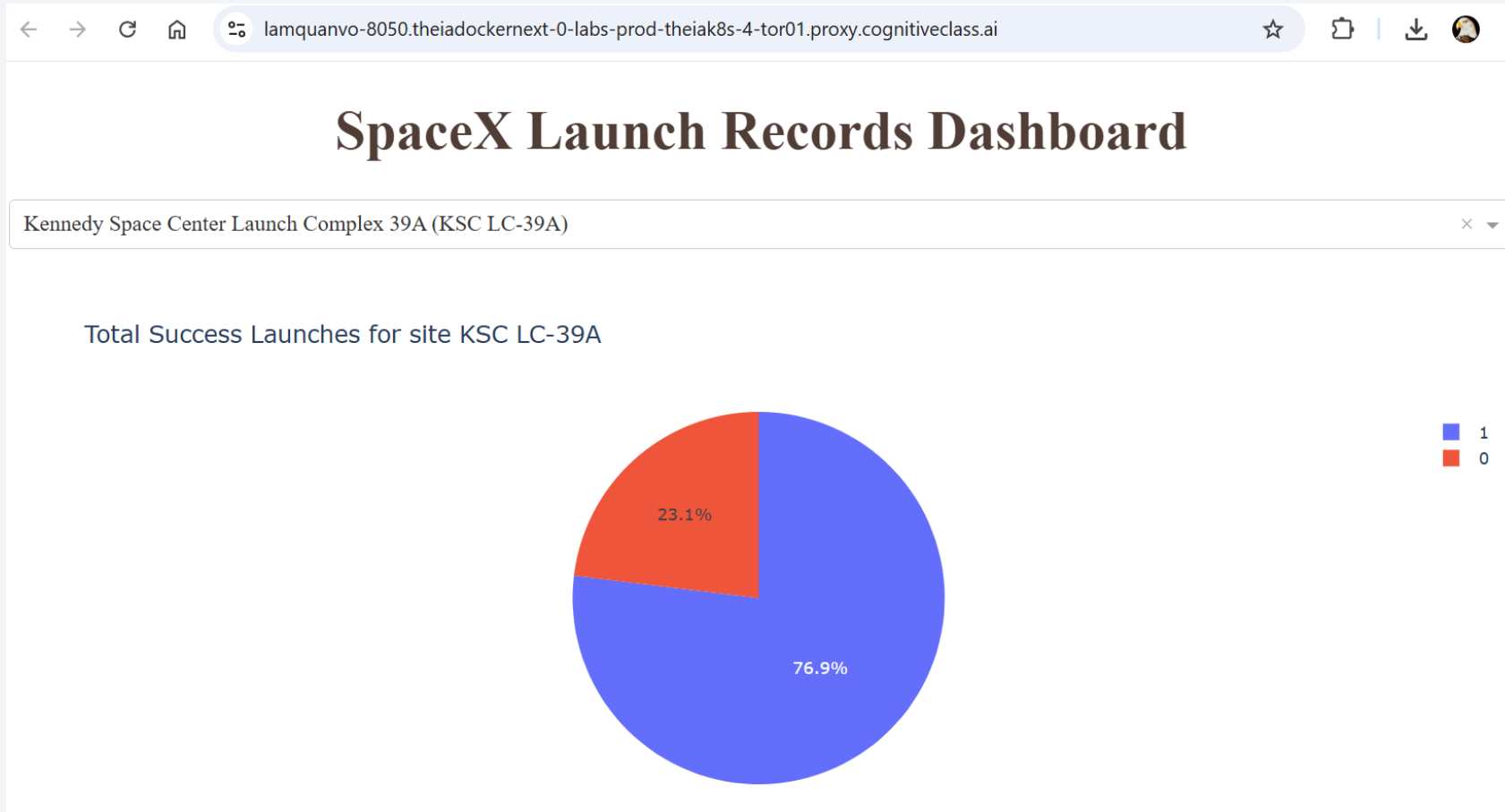
- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

SpaceX Launches by Launch Site



70% of launches from KSC LC & CCAFS LC launch site.

Launches success or fail for site KSC LC-39A



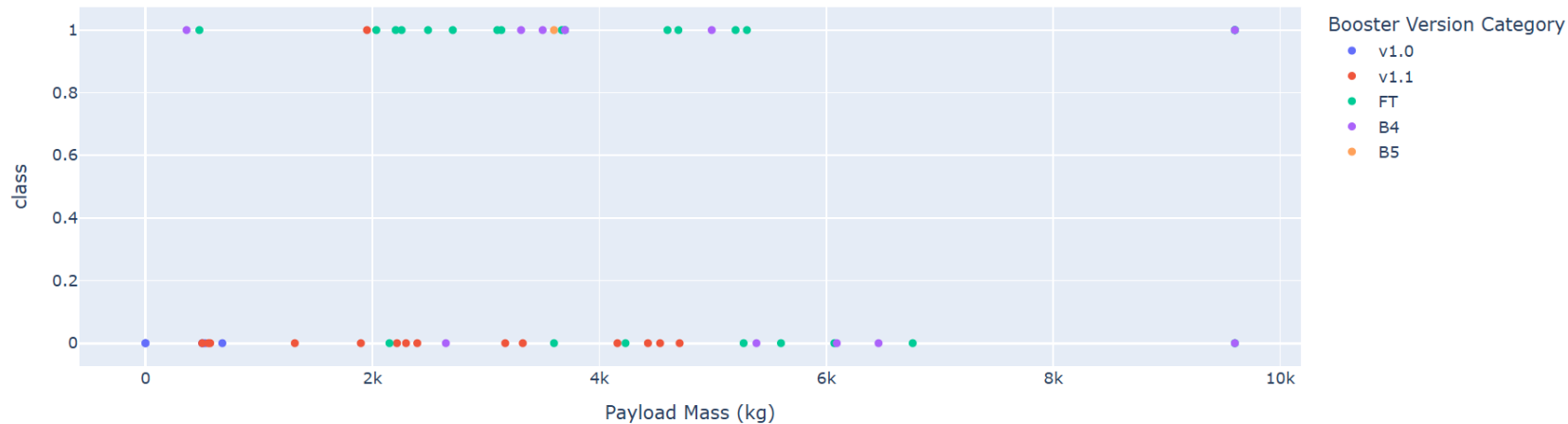
The succeed rate is 3 out of 4 launches.

Launches success or fail by payload mass for all sites

Payload range (Kg):



Success count on Payload mass for all sites

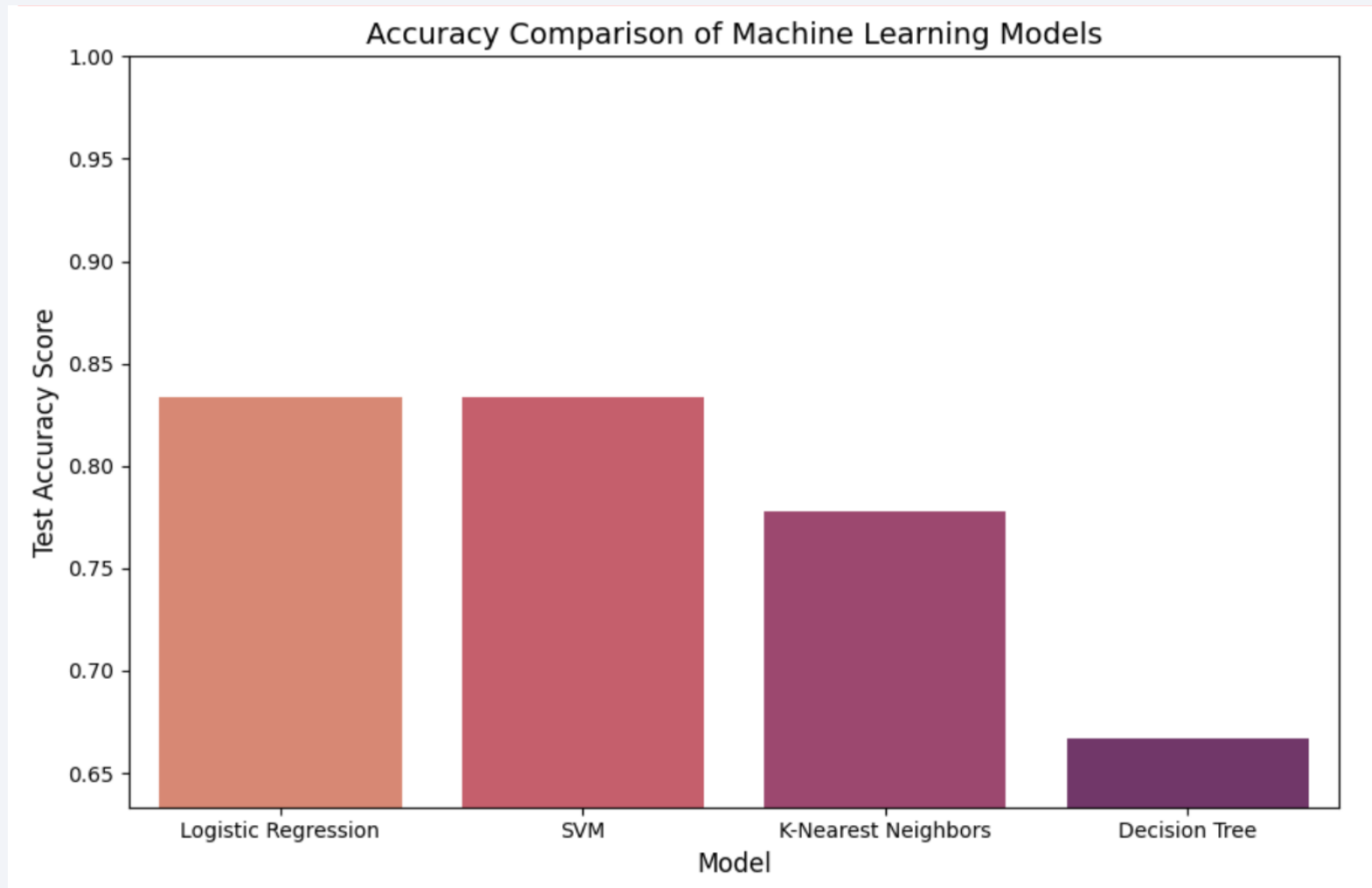


- The range of payload from 2k-4k have the most success launches while the booster version FT is the one with most succeed launches.
- Higher payload more than 6k faces a really big challenges as most fails and only one succeed While the booster version v1.1 is the one with most failures

Section 5

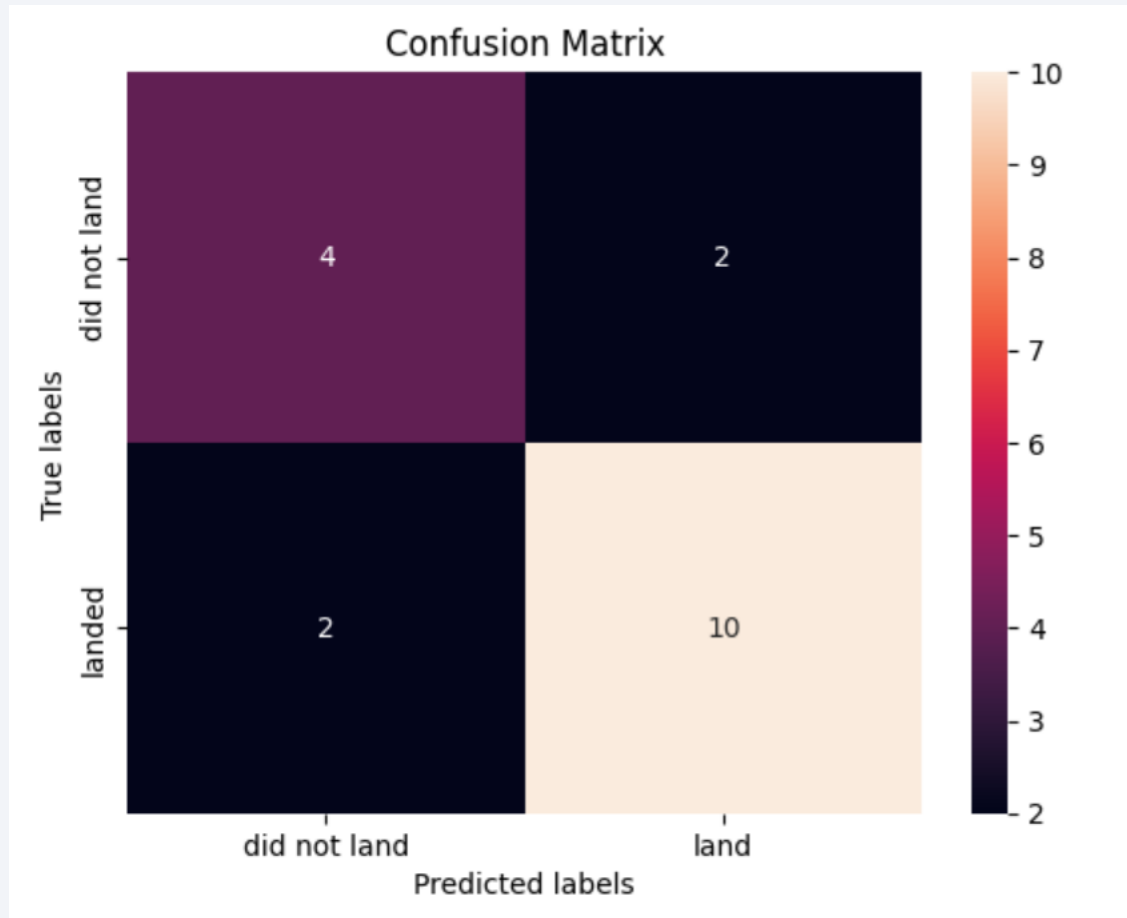
Predictive Analysis (Classification)

Classification Accuracy



- Logistic Regression & SVM have the highest test accuracy score by 0.833333

Confusion Matrix



- Best performing model in this concept is: KNN
 - Accuracy score is only 0.777778
 - Precision score is only 0.833333 (highest)
- Considering the high cost of false positive (predicted the launch is succeed but it is not), I will keen on selecting this model rather than the others.

Conclusions

- There are 4 launch sites across US, and KSC LC & CCAFS LC delivered most of the launches.
- The success ratio kept increasing year over year, indicating that there is the achievement in technology which makes more succeed launches than before.
- KNN is the best performing model with accuracy 78% and precision 83%, considering the high cost of false positives are very significant.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

